

Module 5: Trade, Bids and Auctions

ECO364-THINKING STRATEGICALLY

The Campeau Tender offer

- ▶ Robert Campeau tried to buy the Federated Stores (and its crown jewel, Bloomingdales).
- ▶ He used a two – tiered tender offer.
- ▶ This is a bidding strategy which offers a high price to the first shares tendered and a lower price to the later shares tendered.
- ▶ Let us be more specific.
- ▶ Suppose that: (1) The pre-takeover market price is \$100 per share. (2) The first tier of the bid offers a higher price of \$105 per share up to half of the total shares.
- ▶ The next 50% of the shares (the second tier) are offered a price of \$90 per share
- ▶ (3) But all acquired shares are paid at the same (average) price: if the total share of shares tendered is X , then every share is paid
- ▶ \$105 if $X < 50\%$. If $X \geq 50\%$, then the price paid per share is
- ▶
$$p(X) = \frac{50}{X} 105 + \frac{X-50}{X} 90$$

The Campeau Tender offer

- ▶ For instance, if $X=100$, then
- ▶ $p(X) = \frac{50}{100} 105 + \frac{100-50}{100} 90 = 97.5$
- ▶ If $X=75$, then
- ▶ $p(X) = \frac{50}{75} 105 + \frac{75-50}{175} 90 = 100$
- ▶ If $X=50$, then
- ▶ $p(X) = \frac{50}{50} 105 + \frac{50-50}{50} 90 = 105$
- ▶ Thus, the price per share will range from 97.5 to 105, depending on the percentage of shares acquired

The Campeau Tender offer

- ▶ (4) The two-tiered offer is made unconditional on the number of shares tendered. Even if the raider does not get control (less than 50%), shares are still purchased at \$105 per share – the first tier price
- ▶ (5) If the offer attracts more than 50% of the shares Campeau can force every other share holder to sell his shares for \$90, the second tier price
- ▶ It so happened that in addition to Campeau, Macy's made a conditional offer:
- ▶ It offered \$102 per share provided it got the majority of the shares. In this case every share would be bought at \$102 whether tendered or not
- ▶ To whom should one share holder tender?

Solution

- ▶ **Claim** It is a dominant strategy to tender to the two tiered offer (TTO)
- ▶ **Explanation** Suppose Bob is a shareholder. Consider three cases:
 - ▶ (1) With Bob the TTO attracts less than 50% of total shares
 - ▶ (2) Without Bob the TTO attracts more than 50% of total shares
 - ▶ (3) Bob is pivot, without him the TTO attracts less than 50% and with him it attracts more than 50%
 - ▶ In the first case, if Bob tenders to the TTO he obtains \$105 per share and otherwise he obtains per share either \$100 (if both offers fail) or \$102 (if Macy's offer succeeds). Hence he is best off tendering to the TTO
 - ▶ In the second case, the TTO succeeds and Campeau is the winner whether or not Bob tenders his shares. If Bob does not tender he will obtain \$90 per share while if he tenders he will obtain at least \$97.5 per share.
 - ▶ In the third case, under the assumption that Bob is not a massive share holder (holds less than 12.5%) he will get more than \$102 per share if he tenders to the TTO and will get either \$100 or \$102 per share, otherwise
- ▶ In all three cases Bob is best off tendering to the TTO

Corollary

- ▶ Every shareholder will tender to the TTO and each one of them will obtain \$97.5 per share, less than the current market price
- ▶ This is another example of the Prisoner's Dilemma game, where every player has a dominant strategy but once they use it, the outcome is inferior.
- ▶ Lawyers view the TTO as coercive and have successfully used this as an argument to fight the raider in court.

Remark

- ▶ The conditionality of Macy's offer was key in the second case. If it hadn't been, it is not clear what is Bob's best strategy. If he believed that more than 62.5% of the shares are tendered to the TTO he is best off selling to Macy's!
- ▶ Clearly in such case, to tender to TTO is neither a dominant strategy nor an equilibrium strategy

Equilibrium with Unconditional offer

- ▶ There is no equilibrium where everyone tenders to TTO, and so each share fetches \$97.5 per share. Bob has an incentive to deviate and offer his shares to Macy's in which case he obtains \$102 per share (since Macy's offer is unconditional on having a majority of the shares)
- ▶ There is no equilibrium where everyone tenders to Macy's: Bob would have an incentive to deviate, this time to tender to TTO and obtain \$105 (since $X < 50$)
- ▶ But there is an equilibrium where 62.5% tender to TTO and the other 37.5% tender to Macy's. In this case every share obtains a price of \$102 and no one has an incentive to deviate!

Acquisitions

- ▶ Example 1: You represent Company A (the acquirer) which currently considers acquiring Company T (the target) by means of a tender offer
- ▶ The value of T depends on the outcome of a major oil exploration project. In the worse case (a complete failure), Company T will be worth \$0 per share under the current management. In the best case (a complete success) it will be worth \$100 per share. All intermediate cases, with share values between \$0 and \$100, are equally likely
- ▶ By all estimates, Company T will be worth 50% more in the hands of Company A
- ▶ Finally, Company T owners will accept any offer that is greater or equal to its value under its current management
- ▶ You are asked to make an offer for T's shares before the outcome of the drilling project is known
- ▶ But Company T will know the project's outcome when deciding whether or not to accept your offer
- ▶ What price should you offer?

Acquisitions

- ▶ **Solution:** Suppose you bid \$60 per share. If your bid is accepted then T must be worth at most \$60 per share under the current management (otherwise T will reject your offer). Since all values between 0 and 60 are equally likely, on average T's share will be worth \$30 to the current owner and \$45 to you. By bidding \$60 per share you should expect on average a loss of \$15. In general if you bid \$B per share you should expect to lose $B/4$ per share. Why? If your offer is accepted the value of T's share must be in between 0 and B, equally likely. Thus on average it is $B/2$ under the current management. For you this value is 50% higher, namely $3B/4$. You bid \$B per share and obtain, on average, $3B/4$ per share, hence on average you lose $B/4$ per share. Consequently your best bid is zero per share. This exercise produces the winner's curse in which any positive bid yields an expected loss to the bidder
- ▶ You have witnessed the winner's curse!

**“I wouldn’t want to belong
to any club that would
accept me as a member”**

GROUCHO MARX

Groucho Marx appeared to grasp the intuition behind the “winners’ curse”. If a club’s standards were so low as to accept him, he didn’t want any part of it.



Results from an experiment on that example

- ▶ This experiment was run and with 66 students obtaining the same monetary incentives. with MBA students at Northwestern University, with 123 students obtaining no monetary incentive
- ▶ A false argument is runs as follows: The expected value of T is \$50 and this makes it worth \$75 to me. Therefore I should bid somewhere between 50 and 75
- ▶ It fails to take into account the information contained in “acceptance” (under asymmetric information)
- ▶ A correct analysis must calculate the expected value of T conditional on the bid being accepted.

Bids	No Incentive N=123	Incentive N=66
0	9%	8%
1-49	16%	29%
50-59	37%	26%
60-69	15%	13%
70-79	22%	20%
80+	1%	4%

Acquisitions

- ▶ **Example 2:** Company A considers buying Company B which will accept any offer that reflects a fair value
- ▶ Company B is currently undertaking a major project. If the project is a complete failure the fair value of each share of B will be \$40, and if it is a complete success the fair value of a share will be \$100. As before, all intermediate outcomes (any number between 40 and 100) are possible and equally likely
- ▶ Under A's management, B's value per share will be 50% higher than under the current management
- ▶ A makes an offer to acquire B before knowing the outcome of the project but B answers knowing this outcome
- ▶ What price per share should A offer B?

Acquisitions

- **Solution:** Suppose A offers to purchase all the shares of B for a price of b per share. Then B will accept the offer only if the fair value of a share is between 40 and b . Since all numbers in that range are equally likely the expected (average) fair value of a share of B is:

$$v(b) = \frac{b + 40}{2}$$

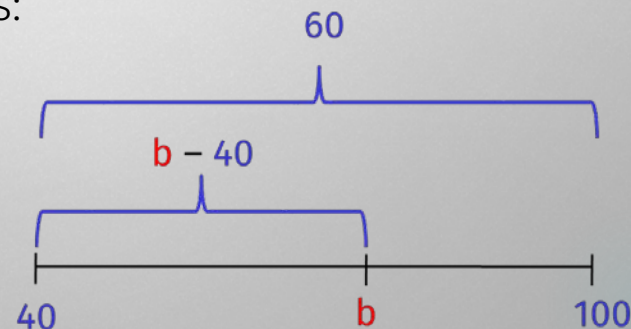
- The expected fair value of a share of B under the skillful management of A is 50% higher. Namely, it is:

$$1.5v(b) = \frac{3(b + 40)}{4}$$

- Hence, conditional on B accepting the offer of A, the expected profit per share for A is:

$$1.5v(b) - b = \frac{3(b + 40)}{4} - b = \frac{120 - b}{4}$$

- But the probability that B will accept the offer b of A is:

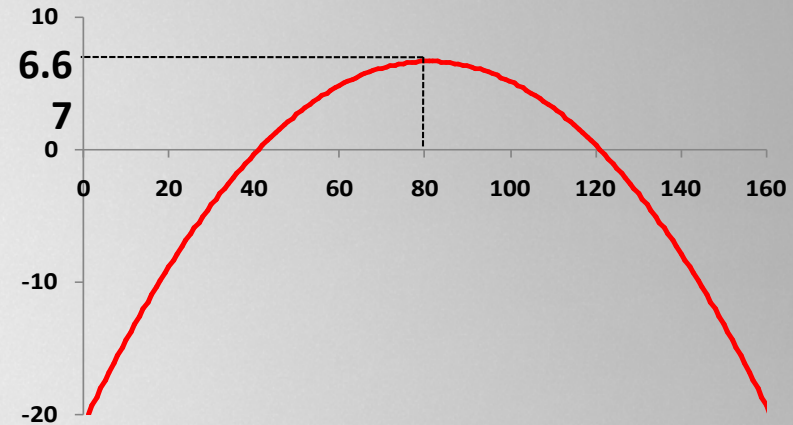


Acquisitions

- Therefore, the expected profit per share for A if he offers b is:

$$\left(\frac{b-40}{60}\right) \cdot \left(\frac{120-b}{4}\right) = \frac{(b-40)(120-b)}{240}$$

- The graph of this profit as a function of b is:



- And it is maximized for $b^* = 80$
- Consequently, A is best off offering \$80 per share of B. On average, A will obtain a profit per share of

$$\frac{(80-40)(120-80)}{240} = \frac{1600}{240} = \$6.66\dots$$

Auctions

- ▶ One of the most common applications of game theory is to auction theory. William Vickery won the Nobel Prize in 1996 mainly for his seminal 1961 paper on auctions.
- ▶ An auction is a strategic game. The players are the seller of an object and the potential buyers (bidders)
- ▶ Typically, the seller chooses the auction format.
- ▶ Then, the strategy of a bidder is how much to bid, or how to bid, depending on the auction format

Auctions

- ▶ Auctions serve as efficient ways to transact all kinds of objects: houses, cars, art, antiques, fresh flowers, livestock, Government bonds, mobile phone licenses, oil drilling rights...
- ▶ (Equivalently, “auctions” are used by one buyer – e.g., the Government– to select a seller and a price (procurement, public contracts,...))
- ▶ The most striking auctions are the ones facilitating the transfer of assets from public to private hands.
- ▶ Auctions eliminate the use of market makers (brokers, specialists), induce competition and usually award the objects to the bidders who value the object the most.
- ▶ Auctions are (typically) simple, quick trading mechanisms, and deal quite effectively with a key problem for the seller:

Ignoring the valuation of potential traders

Some historical notes

- ▶ Auctions have been used (according to Herodotus) in
- ▶ Babylon (500 B.C.). The Babylonians auctioned wives.
- ▶ The ancient Greeks auctioned mine concessions.
- ▶ In 193 A.D. the entire Roman Empire was put on auction by the Praetorian Guard.
- ▶ The winning bid of the winner, Didius julianus, was a promise to pay 25,000 sesterces per man to the Guard.
- ▶ Julianus was beheaded two months after – serving as probably the first and the most extreme example of the winner's curse well known phenomenon.

Auctions and valuations

- ▶ A buyer's valuation is her willingness to pay for the object to be auctioned. (I.e., we say that a buyer's valuation is v if that buyer/bidder is indifferent between paying $\$v$ and obtaining the object and not buying the object)
- ▶ Who knows this piece of information?
- ▶ Think of an auction for a Picasso or some drilling rights for an oil field.
- ▶ Now think of an auction for an apartment.
- ▶ The **maximum willingness to pay is the valuation a bidder assigns to an object.**
- ▶ Namely, if the value is $\$100$ it means that the bidder is indifferent between paying $\$100$ and obtaining the object and not buying the object. He will not pay a penny more for the object.

Auctions: Private values

- ▶ Now think of an auction for an apartment:
- ▶ Each potential buyer (potential resident) knows how much she is willing to pay for it (it depends on her income, tastes, her workplace,...)
- ▶ Knowing how much another potential buyer is willing to pay for the apartment (may be worthwhile but) does not change her own willingness to pay
- ▶ This is the “private values” setting
- ▶ If the factors that determine each bidder’s valuation are not common knowledge and are independent random factors, then we have **independent private values**.

Private values, other alternative settings

- ▶ Non-independent, private values:
 - ▶ My valuation for the apartment depends on whether I obtain a raise in my wage, and this would be more likely if my employer got good reports on profits.
 - ▶ This would also make it more likely that my coworker, who will also bid for the apartment, gets a raise and then has a higher valuation
 - ▶ I got the wage raise...
 - ▶ Then it is highly likely that my coworker also got the raise, and thus her valuation also increased.

Valuations, alternative settings

- ▶ Non private (affiliated) values:
 - ▶ I am bidding for a Picasso
 - ▶ I love the way the subject looks sideways to the peculiar flower while at the same time her mouth seems to send a kiss the other way
 - ▶ But looking at it on my wall is not my only source of pleasure: having a good investment on my wall is another one
 - ▶ In this case, **values are interdependent**. Although they have a private component (my personal enjoyment of the painting), since I also want a good investment, the value of the painting depends on how well it is valued by others in the after-market.
- ▶ Extreme of that: **common values**:
 - ▶ I am bidding for a permit for wildcat drilling
 - ▶ I am a standard oil company (e.g., Standard)
 - ▶ I have conducted some geological studies of the area (as probably have my competitors)
 - ▶ I have private information, but the value for me is just as the value for my competitor: the value of the oil in the patch.
 - ▶ Bidders may acquire different partial information about the value of the oil tract by having access to different geological tests and they will form estimates about this value based on the tests results.
 - ▶ Their bids will depend on these estimates while the bids in the private values case depend only on their values.

Strategies vs. Values

- ▶ The strategy of a bidder is (typically) how much (or how) to bid in the specific auction she participates. This depends on her valuation, if she knows this valuation
- ▶ (Or, more generally, on the information she has about all the valuations.)
- ▶ Valuation and bid (strategy) are two different notions. As a bidder, my valuation of the apartment may be \$350,000 but I may bid
- ▶ (e.g.) only \$275,000

Some Common auction formats

- ▶ We may distinguish between **open bid auctions** and **sealed bid auctions**.
- ▶ Open Bid Auctions
- ▶ **1. Oral Ascending (or English) Auction.**
 - ▶ Beginning at some level, the price of the object is gradually raised until only one bidder remains. This bidder wins the object and pays the last price quoted
 - ▶ (Who raises the price, bidders or auctioneer?)
- ▶ **2. The Japanese Auction.**
 - ▶ Similar to English: The price rises (on a display; a clock) continuously. Bidders keep pressing a button for as long as they desire to remain active, and the auction is active (price keeps rising) as long as at least two bidders are active. Once only one active bidder remains, the auction is over, and that bidder is the winner. She pays the price at which the penultimate bidder dropped out. (Once a bidder drops out, she cannot return.)
- ▶ **3. The Dutch Auction (Oral Descending)**
 - ▶ It works opposite to the English: the auction starts at a very high price and then descends continuously (in a display; a clock). The first bidder to press a button –or call out “Mine”–, stops the clock and wins the object at the price displayed in the clock

Some Common auction formats

- ▶ Sealed Bid Auctions
- ▶ **First-Price, Sealed-Bid Auction (FPA)**
 - ▶ Each bidder submits a bid in sealed envelopes. The envelopes are opened simultaneously.
 - ▶ The highest bidder wins the object and pays her bid.
- ▶ **Second-Price, Sealed-Bid Auction (SPA)**
 - ▶ All as in the FPA, except that the winner does not pay her bid but the second highest submitted bid (highest among rival bids).

Does a second price auction makes sense?

- ▶ Why should the seller forego the difference between the bid “offered” by the winner and the second highest bid?!
- ▶ E.g., the winner offers \$100, the runner-up offers
- ▶ \$75... and the seller charges the winner only \$75!
- ▶ **Wrong question:**
- ▶ The winner was not “offering” \$100; she was “bidding” \$100...
- ▶ She would probably not offer \$100 in a FPA!
- ▶ It is true that in a first price auction you will obtain the highest bid while in the second price auction you will obtain only the second highest bid. But remember, bidders may bid higher in the SPA since they know that if they win they only pay the bid below the winning bid. Indeed in circumstances where bidders are risk neutral and roughly speaking have similar background (together with some other technical assumptions) SPA out performs FPA for the seller.

Bidder behavior: equilibrium in auctions. The private values case

- ▶ Second price auction:
- ▶ Bidders are players in a game
- ▶ Each bidder's strategy set is the set of all (positive) real numbers (or perhaps the set of natural numbers, in cents)
- ▶ Payoff:
 - ▶ If the bidder wins and has valuation v , then $v - b_r$ where b_r is the bid of the runner up
 - ▶ If she does not win, then payoff is 0
 - ▶ With two bidders, a very long and wide table, similar to the ones we are used to handle
 - ▶ But we need to talk about what bidders know!
 - ▶ We have been assuming that the “Table” was common knowledge: in particular, the payoffs (including the rival's)
 - ▶ For this auction format, it does not matter whether this is so: each bidder's (weakly) dominant strategy is to bid
 - ▶ $b = v$

Bidder behavior: equilibrium in auctions. The private values case

- ▶ How about an oral ascending auction (English)?
- ▶ Quite complex, if we were to write the strategy set, etc.
- ▶ Instead, consider the Japanese version, with only two bidders:
- ▶ All a bidder has to decide is when to drop out (can do so at any number)
- ▶ The payoff is $v - b_r$ when she wins and 0 otherwise
- ▶ Same table as the SPA!
- ▶ And yes, same dominant strategy!

Bidder behavior: equilibrium in auctions. The private values case

► The FPA and the Dutch auction

In both cases, the set of strategies is also the set of positive real (or natural) numbers

In both cases, the payoffs are:

- $v - b_o$ when she wins and 0 otherwise, where b_o now
- is the bidder's own bid (or price at which the bidder decides to press the button, in the Dutch auction)
- **Thus, they are really the same GAME**
- **And so have the same equilibrium!**

Bidder behavior: equilibrium in auctions. The private values case

- ▶ The FPA and the Dutch auction

But, what is the equilibrium?

For the time being, it is NOT to bid v !

- ▶ •That is (weakly) dominated by bidding, for instance $v-1$
- ▶ **Main lesson: you change the rules, bidders change their behavior**
- ▶ So, perhaps it is not so stupid to use a SPA: the winner pays less than her bid, but perhaps everybody bids more!

The complete information case

- ▶ Suppose that the valuations of n bidders for a given object are $v_1 > v_2 \geq v_3 \geq \dots \geq v_n$
- ▶ Suppose that these valuations are commonly known. In particular, every bidder knows not only her **valuation** but also the **valuations** of all other bidders
- ▶ Let us examine the equilibrium strategies in first, second and **third** price auctions with the (simplifying) assumption that bids must be integer numbers.

The Complete Information Case

- ▶ Example: Consider four bidders ($n=4$) whose valuations are
- ▶ $v_1=100$, $v_2=90$, $v_3=50$ and $v_4=40$
- ▶ First price Auction
 - ▶ Consider several possible combinations of bids (1) $b_1=95$, $b_2=94$, $b_3=45$ and $b_4=35$
 - ▶ These bids constitute an equilibrium point, but a non-sensible one;
 - ▶ Bidder 2 uses a weakly dominated strategy. She can either obtain zero or lose 4, bidding any number below her valuation is a better strategy for her

The Complete Information Case

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 - ▶ These bids constitute an equilibrium point, but a non-sensible one;
 - ▶ Bidder 2 uses a weakly dominated strategy. She can either obtain zero or lose 4, bidding any number below her valuation is a better strategy for her
 - ▶ (2) $b_1=90$, $b_2=88$, $b_3=45$ and $b_4=35$
 - ▶ This is not an equilibrium point, since bidder 1 is better off reducing her bid to 89 and still winning the auction

The Complete Information Case

- ▶ Example: Consider four bidders ($n=4$) whose valuations are
- ▶ $v_1=100$, $v_2=90$, $v_3=50$ and $v_4=40$
- ▶ First price Auction
 - ▶ Consider several possible combinations of bids
 - ▶ (3) $b_1=88$, $b_2=87$, $b_3=45$ and $b_4=35$
 - ▶ This is not an equilibrium point, since bidder 2 is better off outbidding bidder 1 by bidding 89. In this case her payoff will increase from 0 to 1
 - ▶ (4) $b_1=91$, $b_2=90$, $b_3=45$ and $b_4=35$
 - ▶ This is an equilibrium point, but not w -sensible. Again, bidder 2 uses a w -dominated strategy (she obtains zero even if she wins, any bid below 90 dominates her bid of 90)
 - ▶ (5) $b_1=90$, $b_2=89$, $b_3<50$ and $b_4<40$
 - ▶ This constitutes the unique sensible equilibrium outcome. Note that here every bidder bids below her valuation

The Complete Information Case

► First price Auction:

Summary: Suppose that

$$v_1 > v_2 \geq v_3 \geq \dots \geq v_n$$

Then in a first-price (or Dutch) auction

(i) Every sensible Nash equilibrium must satisfy:

$$b_1 = v_2, b_2 = v_2 - 1, b_3 < v_3, \dots, b_n < v_n$$

(ii) Every Nash equilibrium (sensible or not) is of the form:

$$b_1 = x, b_2 = x - 1, b_3 \leq x - 1, \dots, b_n \leq x - 1 \quad \text{where } v_2 \leq x \leq v_1$$

The Complete Information Case

- ▶ Example: Consider four bidders ($n=4$) whose valuations are
- ▶ $v_1=100, v_2=90, v_3=50$ and $v_4=40$
- ▶ Second-Price (or English) Auction.
 - ▶ First recall that in a SPA it is a w-dominant strategy for every bidder to bid her valuation.
 - ▶ Hence the only sensible equilibrium is
 - ▶ $b_1=v_1, b_2=v_2, \dots, b_n=v_n$
 - ▶ But there are many more non-sensible equilibria.
 - ▶ The combination
 - ▶ $b_1=95, b_2=90, b_3=50, b_4=40$
 - ▶ is also an equilibrium with the same outcome as the sensible equilibrium, namely bidder 1 wins and pays $v_2=90$
 - ▶ There are even other equilibria where bidder 2 wins!
 - ▶ $b_1=80, b_2=105, b_3=50, b_4=40$
 - ▶ This is an equilibrium where 2 wins, pays 80 and obtains a payoff of $90-80=10$. If bidder 1 outbids bidder 2, she will win but will have to pay 105 (more than her valuation)
 - ▶ This equilibrium is not *sensible* since 2 bids above her valuation (a weakly dominated strategy) and 1 bids below his valuation (also a weakly dominated strategy)

Third price Auction: Complete information

- ▶ The rules of this (*peculiar*) auction are as in a FPA, but the winner pays the third highest bid
- ▶ As in the SPA, one can easily argue that bidding below v is (weakly) dominated. Hence in every sensible equilibrium

$b \geq v$
- ▶ Back to our example: $v_1=100, v_2=90, v_3=50$ and $v_4=40$
- ▶ Consider the following bid combination: $b_1=100, b_2=90, b_3=50, b_4=40$
 - ▶ This is not an equilibrium since 2 is better off outbidding!!! She will then win the object for the price of 50
- ▶ For the same reason the following combination is not an equilibrium: $b_1=115, b_2=110, b_3=60, b_4=50$
- ▶ We conclude that to support an equilibrium where 1 wins, bidder 3 should bid at least v_2 (90 in our example)
- ▶ The following combination: $b_1=115, b_2=110, b_3=93, b_4=60$, is an equilibrium, and a sensible one
- ▶ We can change the bid b_3 of 3 to any number in between v_2 and v_1 (in our example, $90 < b_3 < 100$) and the bids will constitute a sensible equilibrium

Third price Auction: Complete information

- ▶ In general any sensible equilibrium outcome in a third price auction must satisfy:

$$(1) \ b_i \geq v_i$$

$$(2) \ b_1 \geq b_2 \geq b_3 \geq b_j, \quad j = 4, \dots, n$$

$$(3) \ v_2 \leq b_3 \leq v_1$$

- ▶ The winner is Bidder 1 who pays the seller between v_2 and v_1 for the object
- ▶ Any number between v_2 and v_1 can be supported as a (sensible) equilibrium price (i.e., as the seller's revenue)
- ▶ For instance, the seller obtains the revenue of 93 in the following sensible equilibrium bid combination:

$$b_1=115, b_2=110, b_3=93, b_4=60$$

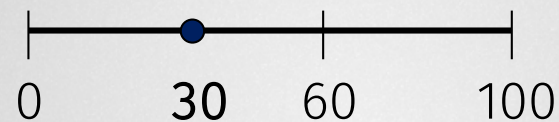
- ▶ Finally, note that there are (non-sensible) equilibria where Bidder 2 is the winner and the seller obtains (much) less than 90:
- ▶ Consider the following bid combination: $b_1=55, b_2=110, b_3=105, b_4=50$
- ▶ Here Bidder 2 wins and pays 55 to the seller. If 1 outbids 2, she wins but pays 105, over his valuation. Note that 1 uses a w-dominated strategy as she bids below her valuation

Going back to incomplete information

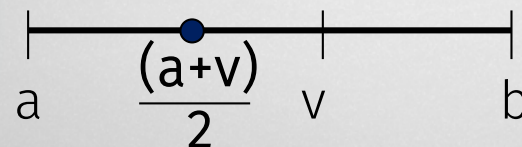
- ▶ The optimal bidding strategy in a First Price Sealed Bid Auction is exactly the same as in the Dutch Descending Auction.
- ▶ In these two auctions bidders do not have a dominant bidding strategy and the equilibrium strategy is very complicated to compute as it depends (unlike the previous case) on the beliefs of the bidders about the other bidders' valuations of the object.
- ▶ Actually, the general bidding strategy is: **BID YOUR ESTIMATE (the expected value) OF THE SECOND HIGHEST VALUATION AMONG THE BIDDERS, ASSUMING YOURS IS THE HIGHEST.**
- ▶ But how to compute this estimate?

Incomplete information: First Price Auction

- ▶ **Example:** Suppose that there are two ($n=2$) bidders and it is commonly known that their valuations of the object is in between 0 to 100, equally likely. Namely, every bidder knows his own valuation and does not know the valuation of his rival. But he knows that it is a number in between 0 and 100 and all the numbers there have the same probability to occur (uniform distribution). This reflects on the fact that a bidder has no prior information about the taste of his rival. Suppose that a bidder has valuation of 60. How much should he bid? By the above rule, he should assume (technically) that his valuation is the highest. Namely his rival's valuation is in between 0 and 60, equally likely



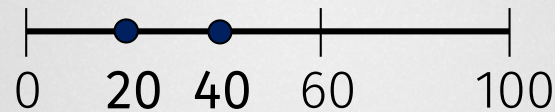
- ▶ Then the expected value of his rival must be 30 and he therefore should bid $b(60)=30$. Similarly, if the bidder's valuation is V he should bid $b(v)= \frac{1}{2}v$.
- ▶ Suppose next that bidder's value for the object is to be in between a and b (instead of 0 to 100), equally likely.



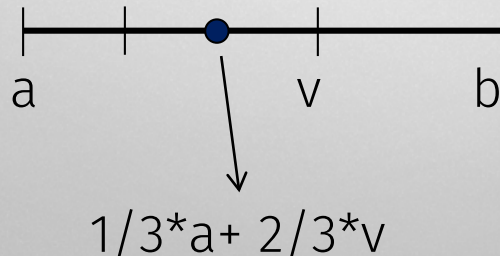
- ▶ Then a bidder with valuation v should bid $b(v)= \frac{1}{2}a+ \frac{1}{2}v$

Incomplete information: First Price Auction

- Suppose next that there are $n=3$ bidders and their valuations are in between 0 and 100, equally likely. A bidder with a valuation of 60 when assuming that his valuation is the highest knows that the valuations of the other two bidders are in between 0 and 60 equally likely. The expected value of the highest of the two is 40 (and the lowest of the two is 20). He therefore should bid $b(60)=40$.



- If the valuations of the three bidders are in between a and b , equally likely than a bidder whose valuation is v should bid $b(v) = \frac{1}{3}a + \frac{2}{3}v$



Incomplete information: First Price Auction

- ▶ The general case: Suppose that there are n bidders and their valuations are commonly known to be in between a and b , equally likely. Then in a First Price Sealed Bid Auction (or in a Dutch Auction) a bidder with valuation v should bid:
- ▶
$$b(v) = \frac{1}{n}a + \frac{n-1}{n}v$$
- ▶ Note that the bid increases with n and it approaches v when n is large, meaning that if there are sufficiently many bidders the winner will pay an amount which is very close to his valuation. The seller has the incentive to increase the number of bidders n , since his payoff increases in n . Also note that the auction is efficient in the sense that the bidder with the highest valuation wins the auction.
- ▶ Example :There are 4 bidders and their valuations are all in between 100 and 1,000, equally likely. What should be the bid of a bidder with valuation of 700, in a FPA?
 - ▶ Answer: $n=4$, $a=100$, $b=1,000$ and $v=700$
 - ▶
$$b(700) = (1/4)*100 + (3/4)*700 = 550$$

Incomplete Information

- ▶ REMARK: Bidding your valuation while it is a dominant strategy in the ascending open auctions or in the second price auction.
- ▶ It is a weakly dominated (non-sensible) strategy in the first price auction or in the Dutch auction.
- ▶ In the latter two auctions a bidder guarantees to obtain zero with this strategy whether or not he wins the auction.

Risk-averse bidders

- ▶ As a seller do you wish to confront risk lover, risk neutral or risk averse – bidders?
- ▶ If your answer is risk lover bidders you are wrong. Risk averse – bidders bid more aggressively in a first price auction (or in the Dutch auction) than risk neutral or risk lover – bidders.
- ▶ They raise their bids to ensure higher probability of winning the object. The risk here is not to win the object.
- ▶ This is in contrast to the second price auction (or English, Japanese or On-line auctions) where bidding truthfully (your valuation) is a dominant strategy for any type of bidders, irrespective of their risk attitude.

THE SELLER'S REVENUE-AUCTION COMPARISON

- ▶ Assuming that all bidders are risk neutral together with some other standard assumptions it can be shown that for the same number of bidders in each auction:
 - ▶ The English Ascending Auction OUTPERFORMS the Second Price Sealed Bid Auction.
 - ▶ The Second Price Sealed Bid Auction OUTPERFORMS the First Price Sealed Bid Auction.
 - ▶ The First Price Sealed Bid Auction IS EQUIVALENT to the Dutch Descending Auction
 - ▶ That is, if you believe that the bidders on e-Bay are risk neutral it is a good idea to sell your item on their site.
 - ▶ On-line auctions are equivalent to the English Auctions.
- ▶ The above comparison is valid for both: private and common value cases.
- ▶ HOWEVER If the bidders are risk averse then the first price auction in certain scenarios (independent valuations) yields the seller higher revenue than the second price auction.
 - ▶ The reason is that bidders bid more aggressively in the former auction.
 - ▶ Also if the seller is risk averse and the bidders are risk neutral again the seller may benefit more in the first price auction. Than in the second price auction.

Issues with auctions

► Entry deterrence

- As it is stated above the English Auction in many circumstances is the best auction for the seller. However, this result is based on the assumption that the same number of bidders participate in all auctions. Under comparison in reality the English Auction may deter entry. For instance, in the auctions for mobile phone licenses, non incumbent firms with even a small disadvantage will see no point entering an open ascending auction.
- Preparation for a large scale auction is usually quite costly and whatever sensible bid a non-incumbent firm makes, it is likely that an incumbent firm will outbid it. A non-incumbent firm may be trapped into winner's curse if she wins in an English Auction against incumbent firms.
- To ease this problem Klemperer in 1998 suggested the Anglo – Dutch Auction. For example Anglo-Dutch Auction for three licenses the price rises continuously until, say, five bidders remain (the English stage).
- Then these five bidders submit sealed-bids with a minimum price which is the one cleared the English stage.
- The highest three bidders win the license and pay the third highest price.
- Here “weak” bidders have incentives to enter such auction since they have a reasonable chance to win the sealed bid competition if they can be among the five winners of the English stage.

Issues with auctions

► Collusion

- Explicit collusion (communication) is often illegal in oral auctions (anti-trust,...), but oral auctions may facilitate tacit collusion
- A simple example: two identical units will be sold in an “English” auction to two potential buyers, who each values having one unit at \$100, and having a second one at additional \$90
- English auction here means: price is increased continuously and bidders decide when to “reduce their demand” –first to 1, then to 0–
- Auction finishes when demand is 2. The units sold at the price at which that happens.
- (One) equilibrium: Each bidder reduces demand to 1 unit when the price reaches 90, then to zero when the price reaches 100
- Outcome: Each bidder obtains one unit and pays \$90...
- Except that this is not the most sensible one. Indeed, bidder 1 may immediately reduce her demand to 1 (at price equal to 0)
- Bidder 2 then knows that it is dominant for bidder 1 to stay put until the price gets to 100!
- So better to also reduce demand to 1 at price 0: each bidder gets one unit at price 0!

Issues with auctions

► Collusion in the real world

- Germany, 1999. Ten blocks of spectrum (licenses) were sold in an ascending auction:
 - Bids on each specific license (paired frequencies)
 - Rounds of simultaneous bids on all
 - Activity rule: A firm could bid on at most as many licenses as the number of valid bids (above the standing price) it bid in the previous round
 - Minimum increment of 10%
 - Four firms participated in the auction: Two big players (Mannesman, and T-Mobile, a subsidiary of Deutch Telecom) and two smaller players (Viag Interkom and E-Plus)
 - This is what happened:

Bidding history in the German GSM auction

Lic #	1	2	3	4	5	6	7	8	9	10
Rd\										
1	36.36 M	36.36 M	36.36 M	36.36 M	36.36 M	40 M	40 M	40 M	40 M	40 M
2	40.01 T	40.01 T	40.01 T	40.01 T	40.01 T	40 M	40 M	40 M	40 M	40 M
3	40.01 T	40.01 T	40.01 T	40.01 T	40.01 T	40 M	40 M	40 M	40 M	40 M

- Round 1: M's jump bid on all licenses
- Round 2: T bids higher on 5 licenses; the rest withdraw
- Round 3: No bids; auction is over

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1	36.36 M	36.36 M	36.36 M	36.36 M	36.36 M	40 M	40 M	40 M	40 M	40 M
2	40.01 T	40.01 T	40.01 T	40.01 T	40.01 T	40 M	40 M	40 M	40 M	40 M
3	40.01 T	40.01 T	40.01 T	40.01 T	40.01 T	40 M	40 M	40 M	40 M	40 M

Source: Grimm, Riedel, and Wolfstetter, IJIO 2003

- Round 1: M's jump bid on all licenses
- Round 2: T bids higher on 5 licenses; the rest withdraw
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- The first one, Mannesman, put first a bid per Megahertz of 18.18m DM on the first five blocks and 20m DM on the second five blocks.
- The second company T-Mobil interpreted the first bid of 18.18m as an offer by Mannesman: Live and let live. Namely, raise our bid of 18.18m on the first five blocks by 10% and bid 20m, but do not outbid us on the second five blocks.
- T-Mobile understood the signal well and indeed they won the first five blocks for 20m DM while Mannesman won the second five blocks also for the same price.

Common-value auctions and the winner's curse

- ▶ We now change the setting to common values (e.g., auctions for –wildcat-mineral rights)
- ▶ All bidders have the same valuation for the “object”, but they do not know what this is. They may have some (imperfect) information related to this
- ▶ For example, When licenses to drill for oil in undersea tracts are put on auction, the amount of oil tract is the same for everybody.
- ▶ But the buyers' estimates of how much oil is likely to be in a tract will depend on their differing geological surveys.
- ▶ Such surveys are not only expensive, they are also very unreliable.
- ▶ Some buyers will therefore receive optimistic surveys, and other will receive pessimistic surveys.

Common-value and the winner's curse

- ▶ For the moment, suppose that the most optimistic buyer will outbid the rest, and so win the license
- ▶ Before the auction, if this buyer had to give the best estimate of the value, she would count only on her information (survey). Say, this estimate is \$100M
- ▶ But, what should she think after the auction, when the official tells her: "Congratulations, you won!"
- ▶ Now she knows not only the result of her survey, but also that everybody else's survey was more pessimistic
- ▶ That is, winning the auction conveys bad news with respect to the amount of oil!

Common-value and the winner's curse

- ▶ A failure to ignore this effect and take it fully into account when bidders decide on their bidding strategy may result with the outcome that the winner pays more than the true value. This happens often in practice. In the example above, suppose that many oil companies compete for the drilling rights to a certain tract of land. Each company obtains an estimate of the value of this tract (the number of oil barrels that can be extracted times the oil price). On average these estimates are likely to be close to the true value.
- ▶ But every firm observes only its estimate and not that of the other companies. Given the difficulty in estimating the amount of oil underground the estimates may vary substantially. Even if companies bid say 15% below their estimate, in the presence of many companies it is likely that the firm whose expert provides the highest estimate wins the auction but loses money.
- ▶ The news that your estimate is the highest is worse the larger is the number of competitors. Thus, the magnitude of the winner's curse increases with the number of bidders in the auction.
- ▶ **BIDDERS NEED TO SHADE THEIR BID WELL BELOW THEIR INITIAL ESTIMATES TO AVOID THE WINNER'S CURSE.**

An experiment with the Winner's Curse

- ▶ A jar full of coins was auctioned off to MBA students at Boston University using a First Price Auction.
- ▶ Unknown to the students, the jar had \$8 worth of coins. A total of 48 such auctions were conducted, 4 in each of 12 classes. No feedback was provided until the entire experiment was completed. Students were also report their estimate of the value of each jar.
- ▶ The mean estimate of the value of the jars was \$5.13 (below the true value: students were not good at estimating the value, but were on average too pessimistic, in fact!)
- ▶ Nevertheless the mean winning bid was \$10.01: winners lost
- ▶ \$2.01 on average!

Winner's curse in Real Life: Field data evidence on bidding for oil and gas drilling rights

- ▶ Consider the Gulf of Mexico leases for 1223 leases issued between 1954 and 1969:
 1. Firms suffer an average present value loss of \$192,128 per lease using 12.5% discount rate.
 2. 62% of all leases were dry. Consequently the winners had no revenues at all.
 3. 16% of all leases were unprofitable although some production occurred.
 4. The other 22% of all leases were profitable and these leases earned only 18.74% rate of return
- ▶ These results appear to reflect excessive enthusiasm for the amount of oil likely to be found.
- ▶ Data surprisingly suggest that bidders make systematic errors.
- ▶ The rationality assumption in economics turns out to be inconsistent with bidders' behavior.

Winner's Curse: A simple mathematical example

- ▶ Two contractors, $i=1,2$, are bidding for a construction project that costs c , a cost which is unknown for both of them.
- ▶ A bid here represents the price at which the contractor is willing to perform the project.
- ▶ They know that cost is equal to $c=s_1 + s_2$
- ▶ s_i is privately known to i , and is uniform in $[0,1]$
- ▶ Knowing $s_1 (=0.6)$, the expected value of c is $s_1 + (1/2)=1.1$
- ▶ But when bidding b , contractor 1 should consider that it will win (and make $b-c$ profits) only if $s_2 > s_1$
- ▶ Conditioning on this, the expected value of c is $s_1 + E[s_2 | s_2 > 0.6]=1.4$
- ▶ If the contractor does not take this into account, winning may be very bad news!

The Max-Bid Auction (just for fun)

- ▶ A car worth \$35,000 is put for sale in the following auction:
 - ▶ The maximum (sealed) bid is \$5,000 and every bidder is allowed to submit as many bids as he wishes.
 - ▶ But once the auction receives 4,000 bids, it closes
 - ▶ Bids are opened; bids that tie with other bids are discarded; the winner is the bidder who put the highest bid among the remaining ones
 - ▶ E.g., if there are two or more bids of \$5,000 then these bids are crossed out, if there are two or more bids equal \$4,999 these bids too are crossed out...
 - ▶ If there is only one bid of \$4,998 then the bidder who placed that bid wins the car and pays \$4,998.
 - ▶ Furthermore, a bidder who wishes to buy a number of bids has the right to do so only if he can place all of them.
 - ▶ The seller is not a philanthropist: he charges \$10 per bid. (Remember: bidders can submit as many bid as they wish until 4,000 bids are received.)
 - ▶ **The seller may extract almost \$45,000 (a profit of \$10,000) and a bidder who wins the car will pay substantially less than \$35,000**
 - ▶ How?

The Max-Bid Auction: Solution

- ▶ Place 2001 consecutive bids from top down (from 5,000 to 3,000) and you ensure your win. → *If there are already more than 1,999 bids and you can not place 2001 bids- stay out*
- ▶ The other 1,999 bids of the other bidders cannot cover all your bids and you guarantee to have the unique highest bid. Your cost is 20,010 for the 2001 bids plus the winning bid – a total of approximately \$25,000.

Incentives and trading

- ▶ The owner of a high-tech company is trying to develop and market a new computer game
- ▶ If he succeeds, he will make a profit of
- ▶ \$200,000
- ▶ If he fails, he makes nothing.
- ▶ Success or failure depends on the expert employee's effort. With high (H) effort, she will succeed with probability 80% and with low (L) effort that probability drops to 60%
- ▶ The expert can be hired for \$50,000 but for this salary she will exert only the low effort level: the cost of H effort for the expert is equivalent to \$20,000
- ▶ To induce her to exert H-effort level, the owner has to pay \$70,000

	Chance of Success	Expected Revenue	Salary	Expected profit
L-effort	60%	\$120,000	\$50,000	\$70,000
H-effort	80%	\$160,000	\$70,000	\$90,000

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- ▶ Clearly the owner will do better to hire a H – effort expert at the higher salary.
- ▶ But there is a problem:
 - ▶ One can't tell by looking at the expert's working day whether she exerts H – effort or L – effort since
 - ▶ the project may fail in both cases with at least 20% probability.
 - ▶ The owner has to base his reward scheme on something he can observe, namely on success (S) or failure (F). Without it the only sensible outcome is that the owner pays the employee a fixed salary of \$50,000 and she exerts a L- effort.

Incentives

- ▶ Indeed, for the expert, exerting low effort is a dominant strategy when the reward is fixed.
- ▶ Thus, with fixed reward, the owner obtains an expected profit of \$70,000.
- ▶ But both could do better. Supposed that the owner pays her a salary of \$50,000 and a bonus of 15% of the actual profit. Then if she exerts a high effort she will earn an expected amount of :

$$50,000 + 0.15 \cdot 0.8 \cdot 200,000 = \$74,000$$

- ▶ The owner will obtain:

$$0.85 \cdot 0.8 \cdot 200,000 - 50,000 = \$86,000$$

- ▶ and both of them are better off.

Incentives

- ▶ The owner should offer a larger sum upon the S outcome and a smaller sum upon the F- outcome. The difference between the two sums (the bonus for success) should just be sufficient so that the expert expects that a high effort will raise her earnings by \$20,000 (from \$50,000 to \$70,000). Suppose that she is paid x if S and paid y if F. Then if she exerts H – effort she obtains an expected payoff of:

$$H : 0.8x + 0.2y$$

- ▶ and if she exerts L – effort she obtains:

$$L : 0.6x + 0.4y$$

- ▶ The difference is $0.2x - 0.2y$ or $0.2(x - y)$ and it should be equal to \$20,000, namely:

$$0.2(x - y) = \$20,000$$

$$x - y = \$100,000$$

- ▶ Therefore, the bonus should be \$100,000. But what is x and what is y ? It is required that:

1. $0.8x + 0.2y = 70,000$

2. $0.6x + 0.4y = 50,000$

- ▶ Solving this system will yield the solution:

- ▶ Multiply (2) by 0.5 to obtain: (3) $0.3x + 0.2y = 25,000$

- ▶ Subtracting (3) from (1) will yield $0.5x = 45,000$, which means that $x = 90,000$, and $y = -10,000$

- ▶ That is, the owner should pay the employee \$90,000 if the outcome is S and the employee should pay a fine of \$10,000 if the outcome is F. This leaves the owner with an expected profit of:

- ▶ $0.8 * 200,000 - 70,000 = \$90,000$

- ▶ The employee obtains an expected payoff of \$70,000 and both the owner and the employee are better off.

Incentives

	Chance of Success	Expected Revenue	Salary	Expected profit
L-effort	60%	\$120,000	\$50,000	\$70,000
H-effort	80%	\$160,000	\$70,000	\$90,000

- ▶ By using the previous scheme, the owner could make as if he could observe the quality of the effort by direct supervision!
- ▶ This incentive scheme is basically a sale of 50% of the firm to the expert for a price of \$10,000.
- ▶ Her net payoff is therefore \$90,000 if S and -\$10,000 if F. It is of her interest to exert H – effort in order to increase the chance of S.
 - ▶ Let's redo it again assuming that the owner sells 50% of the firm to the employee for a price of \$10,000
 - ▶ By exerting effort the expert makes in expectation: $0.5 \cdot 0.8 \cdot 200,000 - 10,000 = \$70,000$
 - ▶ The owner obtains in this case: $0.5 \cdot 0.8 \cdot 200,000 + 10,000 = \$90,000$
- ▶ There maybe implementation problem with these schemes. First, imposing a fine on an employee may not be legal or the worker may not have a capital to pay the \$10,000.

Incentives

- ▶ If the previous scheme is not possible to implement, then the owner should go as close as possible to the equity-sharing solution.
- ▶ Since the minimum effective bonus to the expert is \$100,000 ($x-y=\$100,000$) the scheme should be that the expert will get \$100,000 if S occurs and nothing if F occurs. Now the worker's expected payoff is \$80,000 if she exerts H effort and the expected profit of the owner falls to \$90,000.
- ▶ The inability to enforce fines or get workers to invest their own capital means that the outcome is less attractive to the owner by \$10,000.
- ▶ The owner could do even better if he offers the employee an upfront salary of \$50,000 and 12.5% of the shares of the firm. In this case if she exerts H – effort she obtains
- ▶ $0.125 \cdot 0.8 \cdot 200,000 + 50,000 = 70,000$
- ▶ enough to induce her to exert the H –effort and he obtains 90,000. However with this scheme he takes a bigger risk of ending losing \$50,000 in case of a failure.
- ▶ With the previous scheme (50-50 shared ownership of the firm) none of them could lose any amount.

The two stereo stores: *a tale in advertising*

- ▶ Crazy Eddie announced: *“We cannot be undersold. We will not be undersold. Our prices are the lowest – guaranteed. Our prices are insane”.*
- ▶ His main competitor, Newmark & Lewis is no less ambitious. With any purchase, you get the store’s *“Lifetime low – price guarantee”*. It promises to *rebate double the difference if you can find a lower price elsewhere*. They wrote: *“If, after you purchase, you find the same model advertised or available for sale for low (confirmed printed proof required) by any other local stocking merchant, in this marketing area, during the life time of your purchase, we N&L will gladly refund (by check) 100% of the difference, plus an additional 25% of the difference, or if you prefer, N&L will give you a 200% gift certificate refund (100% of the difference plus an additional 100% of the difference, in gift certificates)”*.
- ▶ They sound very competitive. They promise to beat the rival’s price and one of them is even committed to refund customers if it is not the cheapest.
- ▶ SURPRISINGLY THIS CAN LEAD TO A PRICE SETTING CARTEL.

The two stereo stores

- ▶ Suppose that the two stores sell the same model of DVD which cost them \$100 wholesale and suppose both CE and N&L are selling it for \$150 a piece.
- ▶ Also suppose that the monopoly price is \$200. This is the price a monopoly will set to maximize its profit.
- ▶ What happens if N&L raises its price to \$160?
- ▶ Effectively it sells a DVD for \$140 (she refunds a customer for double the difference between her price and CE's price).
- ▶ Customers will switch from CE who charges \$150 to N&L who effectively charges only \$140.
- ▶ To avoid losing customers CE has no choice but to raise its price to \$160 as well.
- ▶ At this point N&L will again raise the price say to \$170 and CE will have to follow N&L until they
- ▶ reach the cartel price of \$200.
- ▶ At this point none of them has an incentive to change their price.
- ▶ For instance, CE has no incentive to slightly reduce the price in order to capture the market since it knows that this automatically will enhance a double decrease in price by N&L.

Most favored customer clause

- ▶ This clause asserts that the seller will offer to the favored customers the best price they offer to anyone.
- ▶ That is, if the seller offers down the road a discount to any new customer he has to offer the same discount to all the previous favored customers.
- ▶ This enables to sustain a cartel.
- ▶ It reduces the incentives for price cuts: If all cartel members are committed to this clause, then offering discounts to rivals' customers to attract them away from them becomes too costly: the discounts have to be applied to past, own customers too!
- ▶ A well known antitrust case before the Federal Trade Commission involved Du Pont, Ethyl, and other manufactures of gasoline additives.
- ▶ They were charged with using the most favored customer clause.
- ▶ The commission ruled that this clause has an anticompetitive effect, and forbade the companies from using such clauses in their contracts with customers.
- ▶ It has been challenged also as an instrument for a dominant firm (imposing that clause on its contract with suppliers) to foreclose (block the access of) competitors to its market. E.g., case of DOJ against Blue Cross-Blue Shield of Michigan.